

COURSE CODE/TITLE:	CPE 010	SECTION:	CPE21S1
DATE PERFORMED:	July 7, 2026	DATE SUBMITTED:	July 7, 2026
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ACTIVITY NO. ____ **Seatwork 1.1 Algorithm, Flowchart, Pseudocode**

1. PROCEDURE OUTPUT

What procedures did you perform?

1. Computing the total average of 3 quizzes then displaying its total and average.

Algorithm:

1. Start
2. Declare variables: scores[3], total, average
3. For each quiz (1 to 3):
 Ask user to input a score
 Store it in scores
4. Add up all 3 scores → total
5. Divide total by 3 → average
6. Display total and average
7. End

Pseudocode:

```

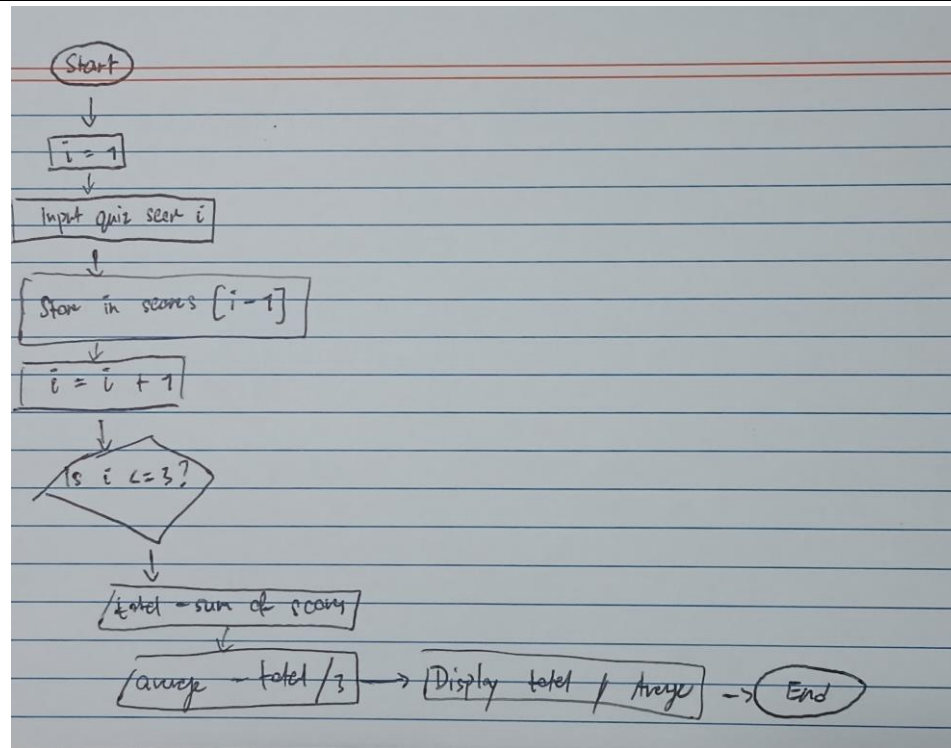
BEGIN
    DECLARE scores[3] AS INTEGER
    DECLARE total, average AS NUMBER

    FOR i = 1 TO 3
        DISPLAY "Quiz #" + i + ": "
        INPUT scores[i-1]
    END FOR

    total = scores[0] + scores[1] + scores[2]
    average = total / 3

    DISPLAY "Total: " + total
    DISPLAY "Average: " + average
END

```



Flowchart

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main.c... **Output**

```
1  #include <iostream>
2
3  using namespace std;
4
5  int main() {
6
7      int x;
8      int scores[3];
9      double average;
10     int total;
11
12     for (int i = 1; i<4; i++) {
13         cout << "Quiz #" << i << ": ";
14         cin >> x;
15         scores[i-1] = x;
16     }
17
18     total = scores[0]+scores[1]+scores[2];
19     average = total/3.0;
20
21     cout << "Total: " << total << endl;
22     cout << "Average: " << average;
23
24     return 0;
25 }
```

Run

C++ Program (Input)

The screenshot shows a mobile browser interface with a dark theme. At the top, the status bar displays the time 10:28, date Tue, Jul 7, and battery level at 24%. The browser address bar shows 'Online C++ Compiler - Programiz' and 'programiz.com'. The page header includes the Programiz logo, 'C++ Online Compiler', and a 'Programiz PRO' button. Below the header, there are tabs for 'main.c...' and 'Output', with the 'Output' tab selected. The output area displays the following text: 'Quiz #1: 10', 'Quiz #2: 5', 'Quiz #3: 10', 'Total: 25', 'Average: 8.33333', and '=== Code Execution Successful ==='. At the bottom of the screen is an Android-style navigation bar with various app icons.

```
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main.c... Output
Quiz #1: 10
Quiz #2: 5
Quiz #3: 10
Total: 25
Average: 8.33333
=== Code Execution Successful ===
```

C++ Program Output

2. Cashiering system that computes for total price of the 3 items bought, accept payment and computes change. Display the transactions made.

Algorithm:

Start

Declare arrays to store transaction history: totalHistory, paid, changeHistory

Set saleCount = 0

Repeat:

Display menu (1. New Transaction, 2. View History, 3. Exit)

Input choice

If choice == 1:

Input price of item1, item2, item3

Compute total = item1 + item2 + item3

Display total

Input payment

While payment < total: ask for payment again

Compute change = payment - total

Display receipt (items, total, payment, change)

Save total, payment, change into history arrays

Increase saleCount by 1

Else if choice == 2:

Display all saved transactions from history arrays

Else if choice == 3:

Exit program

End

Pseudocode:

BEGIN

DECLARE totalHistory[100], paid[100], changeHistory[100] AS NUMBER

DECLARE saleCount = 0

DECLARE choice AS INTEGER

LOOP

DISPLAY "====Cashier System===="

DISPLAY "1. New Transaction (3 items)"

DISPLAY "2. View Transaction History"

DISPLAY "3. Exit"

INPUT choice

IF choice == 1 THEN

DECLARE item1, item2, item3 AS NUMBER

DISPLAY "Enter price of item 1: "

INPUT item1

DISPLAY "Enter price of item 2: "

INPUT item2

DISPLAY "Enter price of item 3: "

INPUT item3

total = item1 + item2 + item3

```
DISPLAY "Total Price: " + total

DISPLAY "Enter payment: "
INPUT payment

WHILE payment < total
    DISPLAY "Payment is not enough. Enter a higher amount: "
    INPUT payment
END WHILE

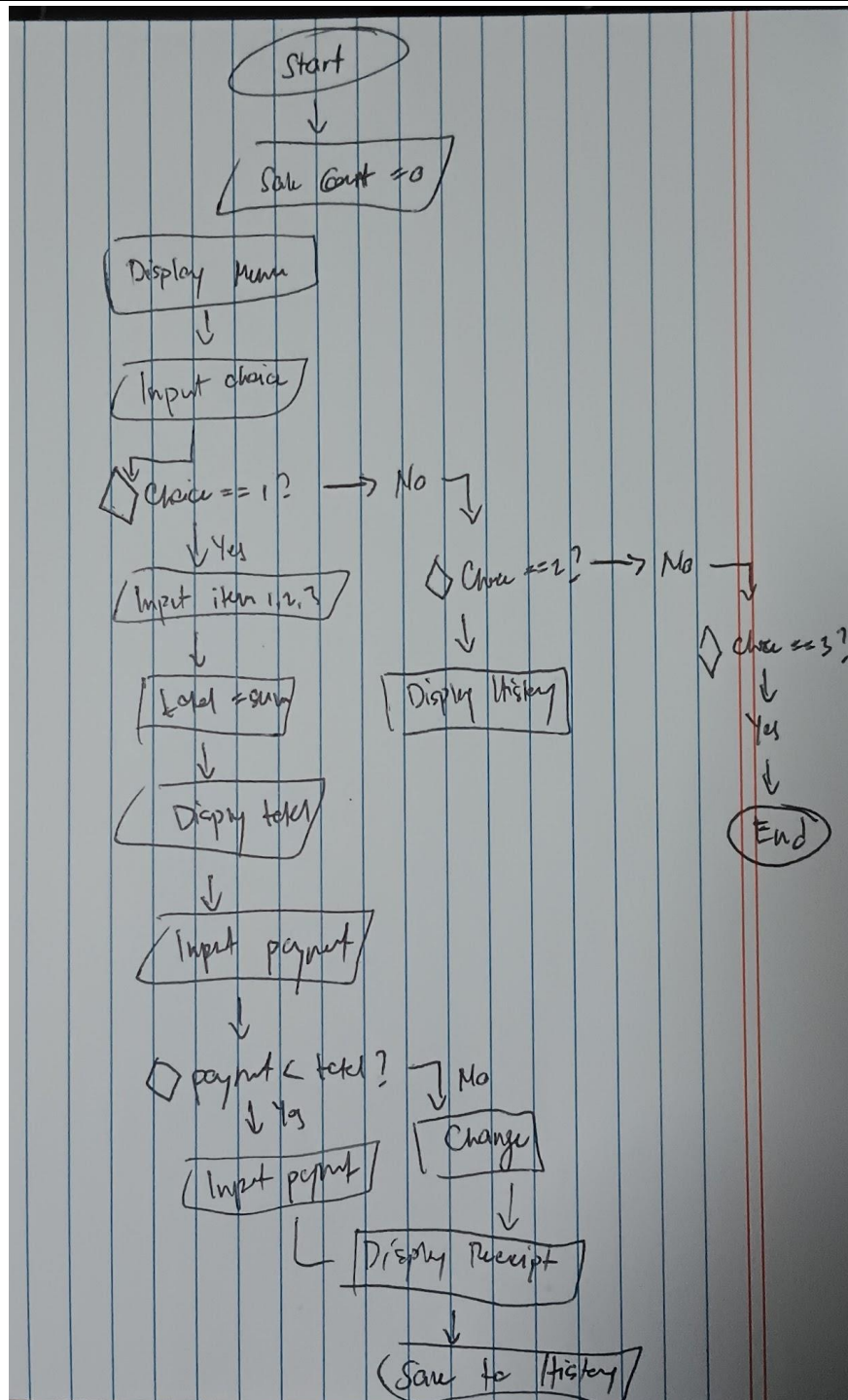
change = payment - total

DISPLAY "*****Receipt*****"
DISPLAY "Item 1: " + item1
DISPLAY "Item 2: " + item2
DISPLAY "Item 3: " + item3
DISPLAY "Total: " + total
DISPLAY "Payment: " + payment
DISPLAY "Change: " + change

totalHistory[saleCount] = total
paid[saleCount] = payment
changeHistory[saleCount] = change
saleCount = saleCount + 1

ELSE IF choice == 2 THEN
    DISPLAY "====Transaction History===="
    FOR i = 0 TO saleCount - 1
        DISPLAY "Transaction #" + (i+1)
        DISPLAY "Total: " + totalHistory[i]
        DISPLAY "Paid: " + paid[i]
        DISPLAY "Change: " + changeHistory[i]
    END FOR

ELSE IF choice == 3 THEN
    DISPLAY "Exiting program..."
    EXIT LOOP
END IF
END LOOP
END
```



Flowchart

```
1 #include <iostream>
2 #include <iomanip>
3
4 using namespace std;
5
6 int main() {
7
8     int maxTransaction = 100;
9     double totalHistory[maxTransaction];
10    double paid[maxTransaction];
11    double changeHistory[maxTransaction];
12
13    int saleCount = 0;
14
15    int choice;
16
17    while(true){
18        cout << "====Cashier System====" << endl;
19        cout << "-----" << endl;
20        cout << "1. New Transaction (3 items)" << endl;
21        cout << "2. View Transaction History" << endl;
22        cout << "3. Exit" << endl;
23        cout << "-----" << endl;
24        cout << "Enter choice: ";
25        cin >> choice;
26
27        if(choice == 1){
28            double item1, item2, item3;
29            cout << "\nEnter price of item 1: ";
30            cin >> item1;
31            cout << "\nEnter price of item 2: ";
32            cin >> item2;
33            cout << "\nEnter price of item 3: ";
34            cin >> item3;
35
36            double total = item1 + item2 + item3;
37
38            cout << "\nTotal Price: " << total << " pesos" << endl;
39
40            double payment;
41            cout << "Enter payment: ";
42            cin >> payment;
43
44            while (payment < total){
45                cout << "Payment is not enough. Please enter a higher
46                    amount: ";
47                cin >> payment;
48            }
49
50            double change = payment - total;
```


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main.c...

Output

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```

double change = payment - total;

cout << "\n*****Receipt*****" << endl;
cout << left << setw(20) << "Item 1:" << item1 << " pesos" <<
endl;
cout << left << setw(20) << "Item 2:" << item2 << " pesos" <<
endl;
cout << left << setw(20) << "Item 3:" << item3 << " pesos" <<
endl;
cout << left << setw(20) << "Total:" << total << " pesos" <<
endl;
cout << left << setw(20) << "Payment:" << payment << " pesos"
<< endl;
cout << left << setw(20) << "Change:" << change << " pesos" <<
endl;
cout << "*****" << endl;

totalHistory[saleCount] = total;
paid[saleCount] = payment;
changeHistory[saleCount] = change;
saleCount++;

} else if(choice == 2){
    cout << "\n====Transaction History====" << endl;
    for(int i = 0; i < saleCount; i++){
        cout << "Transaction #" << (i+1) << " - ";
        cout << "Total: " << totalHistory[i] << " | ";
        cout << "Paid: " << paid[i] << " | ";
        cout << "Change: " << changeHistory[i] << endl;
    }
    cout << "=====" << endl;

} else if(choice == 3){
    cout << "Exiting program..." << endl;
    break;
}

return 0;
}

```

Run

C++ Program (Input)

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main.c... Output

====Cashier System====
-----
1. New Transaction (3 items)
2. View Transaction History
3. Exit
-----
Enter choice: 1

Enter price of item 1: 10

Enter price of item 2: 10

Enter price of item 3: 10

Total Price: 30 pesos
Enter payment: 20
Payment is not enough. Please enter a higher amount: 30

*****Receipt*****
Item 1:      10 pesos
Item 2:      10 pesos
Item 3:      10 pesos
Total:       30 pesos
Payment:     30 pesos
Change:      0 pesos
*****

```

```

====Cashier System====
-----
1. New Transaction (3 items)
2. View Transaction History
3. Exit
-----
Enter choice: 2

====Transaction History====
Transaction #1 - Total: 30 | Paid: 30 | Change: 0
=====
====Cashier System====
-----
1. New Transaction (3 items)
2. View Transaction History
3. Exit
-----
Enter choice: 3
Exiting program...

=== Code Execution Successful ===

```

C++ Program (Output)

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Claude, Gemini.

Purpose of Use:

Checking Syntax if it is correct, searching formal sentences and words.

Extent of Use:

By using claude it helps me to fix my syntax, I usually asked it to "check my code if there is missing, but don't generate a code to solve it, just explain to me what should I do", well because I always start with a scratch of code, like drafting. So that I could master it. Same as Gemini if i already use claude too much that it tells me to avail its subscription plan.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.



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